Radian GS8048 Communications

This document provides a preliminary insight into the new MATE3 communications as regards the upcoming Radian GS8048 inverter, and is intended for advanced programmers intimately familiar with the MATE and MATE3 Serial Communications structure.

JSON Data

{"devstatus": {
"Sys_Time": 1311260338,
"Sys_Batt_V": 26.9,
"ports": [
{ "Port": 1, "Dev": "GS","Type": "60Hz","Inv_I_L1": 0,"Chg_I_L1": 0,"Buy_I_L1": 0,"Sell_I_L1": 0,"VAC1_in_L1": 0,"VAC2_in_L1": 0,"VAC_out_L1": 0,"Inv_I_L2": 0,"Chg_I_L2": 0,"Buy_I_L2": 0,"Sell_I_L2": 0,"VAC1_in_L2": 0,"VAC2_in_L2": 0,"VAC_out_L2": 0,"AC_Input": "Gen","Batt_V": 26.4,"AC_mode": "NO AC","INV_mode": "PassThru","Warn": ["AC In Freq Low","In VAC Low"],"Error": ["none"],"AUX": "enabled","RELAY": "disabled"}]}}

Serial Data Stream

01,6,00,00,00,00,119,000,119,00,00,00,00,120,000,120,10,000,00,264,026,009,CCC = Checksum digits
01,6,00,00,00,00,119,000,119,00,00,00,00,120,000,121,10,000,00,264,026,009,CCC
01,6,00,00,00,00,119,000,120,00,00,00,00,120,000,120,10,000,00,264,026,009,CCC
01,6,00,00,00,00,120,000,119,00,00,00,00,121,000,121,10,000,00,264,026,009,CCC
01,6,00,00,00,00,119,000,119,00,00,00,00,120,000,121,10,000,00,264,026,009,CCC
01,6,00,00,00,00,119,000,120,00,00,00,00,120,000,120,10,000,00,264,026,009,CCC

Byte Format Structure

BYTE	Description
1.	ASCII (10) New Line character denoting the start of the status page.
2.	Tens digit of Port number.
3.	Ones digit of Port number.
4.	ASCII (44) a comma as a data separator.
5.	Device type ASCII (50) '6' GS.
6.	ASCII (44) a comma as a data separator.
7.	Tens digit of L1 Inverter current.
8.	Ones digit of L1 Inverter current.
9.	ASCII (44) a comma as a data separator.
10.	Tens digit of L1 Charger current.
11.	Ones digit of L1 Charger current.
12.	ASCII (44) a comma as a data separator.

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13.	Tens digit of L1 Buy current.
14.	Ones digit of L1 Buy current.
15.	ASCII (44) a comma as a data separator.
16.	Tens digit of L1 Sell current.
17.	Ones digit of L1 Sell current.
18.	ASCII (44) a comma as a data separator
19.	Hundreds digit of the L1 Grid AC input voltage.
20.	Tens digit of L1 Grid AC input voltage.
21.	Ones digit of L1 Grid AC input voltage.
22.	ASCII (44) a comma as a data separator.
23.	Hundreds digit of the L1 Gen AC input voltage.
24.	Tens digit of L1 Gen AC input voltage.
25.	Ones digit of L1 Gen AC input voltage.
26.	ASCII (44) a comma as a data separator.
27.	Hundreds digit of the L1 AC output voltage.
28.	Tens digit of L1 AC output voltage.
29.	Ones digit of L1 AC output voltage.
30.	ASCII (44) a comma as a data separator.
31.	Tens digit of L2 Inverter current.
32.	Ones digit of L2 Inverter current.
33.	ASCII (44) a comma as a data separator.
34.	Tens digit of L2 Charger current.
35.	Ones digit of L2 Charger current.
36.	ASCII (44) a comma as a data separator.
37.	Tens digit of L2 Buy current.
38.	Ones digit of L2 Buy current.
39.	ASCII (44) a comma as a data separator.
40.	Tens digit of L2 Sell current.
41.	Ones digit of L2 Sell current.
42.	ASCII (44) a comma as a data separator
43.	Hundreds digit of the L2 Grid AC input voltage.
44.	Tens digit of L2 Grid AC input voltage.
45.	Ones digit of L2 Grid AC input voltage.

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46.	ASCII (44) a comma as a data separator.
47.	Hundreds digit of the L2 Gen AC input voltage.
48.	Tens digit of L2 Gen AC input voltage.
49.	Ones digit of L2 Gen AC input voltage.
50.	ASCII (44) a comma as a data separator.
51.	Hundreds digit of the L2 AC output voltage.
52.	Tens digit of L2 AC output voltage.
53.	Ones digit of L2 AC output voltage.
54.	ASCII (44) a comma as a data separator.
55.	Tens digit of GS operating mode.
56.	Ones digit of GS operating mode.
57.	ASCII (44) a comma as a data separator.
58.	High byte of GS Error mode.
59.	Middle byte of GS Error mode.
60.	Low byte of GS Error mode.
61.	ASCII (44) a comma as a data separator.
62.	High byte of GS AC mode.
63.	Low byte of GS AC mode
64.	ASCII (44) a comma as a data separator.
65.	Tens digit of GS battery voltage.
66.	Ones digit of GS battery voltage.
67.	Tenths digit of GS battery voltage.
68.	ASCII (44) a comma as a data separator.
69.	High byte of GS Misc.
70.	Middle byte of GS Misc.
71.	Low byte of GS Misc.
72.	ASCII (44) a comma as a data separator.
73.	High byte of GS Warning mode.
74.	Middle byte of GS Warning mode.
75.	Low byte of GS Warning mode.
76.	ASCII (44) a comma as a data separator.
77.	Hundreds digit of Chksum.
78.	Tens digit of Chksum.
79.	Ones digit of Chksum.
80.	ASCII (13) carriage return. Denotes end of status page.

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GS Misc

BYTE	Description
7	0
6	0
5	1= AUX Relay enabled, 0= AUX Relay disabled
4	1= 12V AUX Output enabled, 0= 12V AUX Output disabled
3	1=60Hz, 0 = 50Hz
2	1 = 48V Radian, 0 = 24V Radian
1	1 = 24V Radian, 0 = 48V Radian
0	0

Error Mode (Same as FX)

BYTE	Description
7	Back Feed
6	Shorted Output
5	High Battery
4	Phase Loss
3	Low Battery
2	Over Temp
1	Stacking Error
0	Low VAC Output

Warn Mode (Same as FX)

BYTE	Description
7	Fan Failure
6	Comm Error
5	Temp Sensor Failed
4	Buy Amps > Input size
3	Input VAC Low
2	Input VAC High
1	AC Input Freq Low
0	AC Input Freq High

GS AC Mode (Same as FX)

BYTE	Description
0	No AC
01	AC Drop
02	AC Use

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GS Operating Mode

BYTE	Description
01	Search
02	Inverting
03	Charging
08	Support
09	Selling
10	Pass Through
14	Offsetting
90	GS Error
91	AGS Error
92	Comm Error

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